SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40 **Total Marks:40**

Annexure A

Mock Interview

QUESTIONS:40 **Total Marks:40**

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)

Answer:
Network congestion occurs when data traffic exceeds link capacity or router buffer space, causing queueing delay to increase exponentially. As buffers overflow, incoming packets are dropped, leading to high packet loss. This forces TCP retransmissions, further increasing latency and severely degrading overall throughput.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)

Answer:
In a star topology, nodes connect to a central switch; a single link failure affects only one node, but a switch failure causes network-wide outage (single point of failure). In a mesh topology, redundant point-to-point links allow traffic to be dynamically rerouted if any link fails, providing superior fault tolerance and reliability at higher cabling cost.

3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)

Answer:
Every intermediate router adds processing delay (header inspection/lookup), queueing delay (buffer waiting time), transmission delay (serializing bits onto the wire), and propagation delay over the link. Accumulating these delays across more router hops directly increases total end-to-end Round Trip Time (RTT).

4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)

Answer:
QoS improves voice and video performance by classifying and prioritizing real-time traffic over delay-tolerant traffic. Techniques such as DSCP marking, priority queuing, traffic shaping, and bandwidth reservation reduce queueing delay, jitter, and packet loss during congestion. This helps maintain smoother voice and video communication, although the actual performance depends on available bandwidth and network conditions.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)

Answer:
Fiber optic cables transmit data as light pulses over glass, offering gigabit/terabit bandwidth,

minimal signal attenuation over long distances (tens of kilometers), and complete immunity to Electromagnetic Interference (EMI). Copper cables suffer from higher attenuation beyond 100 meters, EMI susceptibility, and strict throughput limits.

6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)

Answer:
VLANs segment a physical LAN into isolated logical broadcast domains. Performance improves because broadcast/multicast traffic is confined to specific subnets, reducing CPU load on hosts. Security improves by preventing unauthorized inter-VLAN eavesdropping and requiring traffic between VLANs to pass through a firewall/router with strict ACLs.

7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)

Answer:
TCP is connection-oriented, reliable, and provides flow/congestion control with retransmissions, introducing overhead and variable latency—ideal for error-intolerant applications (HTTP/HTTPS, SSH, File Transfer). UDP is connectionless and header-light with no retransmissions, delivering minimal latency—optimal for real-time applications (VoIP, video streaming, DNS, gaming).

8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)

Answer:
Load balancing distributes incoming traffic across multiple servers, firewalls, or WAN links. Impact on performance: Prevents individual hardware bottlenecks, optimizing throughput and response times. Impact on availability: Conducts continuous health checks; if a server fails, traffic is seamlessly rerouted to healthy nodes, eliminating single points of failure.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)

Answer:
Frequent routing updates consume network bandwidth and router CPU cycles. While rapid updates allow faster network convergence after failure, excessively frequent updates cause route flapping, high control-plane overhead, temporary routing loops, and CPU saturation. Link-state protocols like OSPF optimize this by sending triggered updates only upon topology changes.

10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)

Answer:
Security measures like Next-Generation Firewalls (deep packet inspection) and cryptographic processing (IPsec/TLS encryption) introduce computational latency and CPU overhead. If security hardware is undersized, it creates a throughput bottleneck. The trade-off requires hardware ASIC crypto-acceleration and optimized firewall policies to balance strong security with high performance.

11. A network has high delay—how will you identify and fix the issue? (BL4)(1)

Answer:
1. Identify symptom: High latency reported by users or monitoring tools. 2. Measure metric: Run traceroute and ICMP ping to pinpoint the bottleneck hop where RTT spikes. 3. Locate cause: Check interface utilization, buffer queue depth, and CPU load on the bottleneck router. 4. Apply fix: Upgrade link bandwidth, enable QoS prioritization, optimize routing metrics, or fix duplex mismatches. 5. Verify result: Re-test RTT to confirm latency is restored.

12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)

Answer:
Low throughput on a high-bandwidth link is typically caused by: 1. High TCP packet loss causing congestion window throttling. 2. High Round Trip Time (RTT) limiting TCP window scaling. 3. Duplex mismatches causing heavy collisions. 4. Misconfigured TCP window size or MTU/MSS fragmentation. 5. Intermediate device CPU saturation or aggressive rate-limiting policies.

13. How would you reduce packet loss in a congested network? (BL4)(1)

Answer:
1. Implement Quality of Service (QoS) with Random Early Detection (RED/WRED) to avoid buffer tail drop and TCP global synchronization. 2. Increase buffer depth or upgrade saturated link capacity. 3. Configure traffic shaping and policing to smooth traffic bursts. 4. Use link aggregation (LACP/EtherChannel). 5. Offload bulk data to secondary links using policy-based routing.

14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)

Answer:
1. Identify symptom: Audio fragmentation and breakup during VoIP calls. 2. Measure metric: Measure packet loss (>1%), jitter (>30 ms), and delay (>150 ms) using monitoring tools or Wireshark. 3. Locate cause: Check switch/router interfaces for drops or lack of QoS tagging. 4. Apply fix: Enable QoS (mark VoIP with DSCP EF / CoS 5) assigned to Low Latency Queueing (LLQ), and disable SIP ALG on firewalls. 5. Verify result: Conduct test calls to confirm clear audio.

15. Given a network with high jitter, suggest improvements (BL4)(1)

Answer:
High jitter (variable arrival delay) is improved by: 1. Configuring Priority Queueing (PQ) / Strict Priority QoS for real-time packets to ensure consistent queue processing. 2. Enabling jitter buffers on receiving VoIP endpoints. 3. Over-provisioning bandwidth on congested bottleneck links. 4. Isolating voice traffic on a dedicated Voice VLAN.

16. How will you simulate a congested network in Packet Tracer? (BL5)(1)

Answer:
1. Create a topology in Cisco Packet Tracer with several PCs generating traffic through a router or switch connected to a relatively low-bandwidth bottleneck link. 2. Generate continuous traffic using repeated ping or application traffic such as FTP where supported. 3. Use Simulation Mode to observe packet queues, delays, and dropped PDUs. 4. Check router interface statistics using commands such as show interfaces to identify congestion and packet drops. 5. Compare the network behaviour before and during the generated traffic load.

17. If packets are delayed in a router, what steps will you take? (BL4)(1)

Answer:
1. Inspect router CPU and memory utilization using `show process cpu`. 2. Check interface output queue statistics using `show interface` to identify queueing delay and drops. 3. Verify hardware switching mode (ensure Cisco Express Forwarding - CEF is enabled). 4. Apply QoS queuing algorithms (WFQ or LLQ) to prevent low-priority traffic from blocking high-priority packets.

18. How would you optimize a network for online gaming? (BL4)(1)

Answer:
1. Prefer wired Ethernet for gaming systems where possible to reduce wireless interference and jitter. 2. Configure QoS to prioritize gaming traffic and prevent large downloads or uploads from increasing latency. 3. Reduce bufferbloat by using appropriate queue management on the router. 4. Select a stable, low-latency Internet path and avoid unnecessary network hops. 5. Monitor ping, jitter, packet loss, and RTT while testing the connection.

19. Design a small network and explain how you will measure its performance (BL5)(1)

Answer:
Design: A star topology with a Layer 3 Router, Managed Switch, Web Server, and client PCs in separate VLANs (VLAN 10 Data, VLAN 20 Voice). Measurement: Use ICMP Ping/Traceroute for RTT/delay, iperf3 for maximum TCP/UDP throughput, Wireshark for packet loss/jitter analysis, and Packet Tracer Simulation Mode to inspect PDU processing across link hops.

20. If RTT is very high, how will you diagnose the issue? (BL4)(1)

Answer:
1. Execute `traceroute` to identify the exact router hop where RTT increases significantly. 2. Perform ICMP ping tests with varying packet sizes to check for MTU fragmentation or link degradation. 3. Inspect intermediate routers for CPU saturation, buffer congestion, or suboptimal routing paths. 4. Verify physical cabling and WAN interface status for errors.

21. Explain network performance to a non-technical person (BL5)(1)

Answer:
Imagine a network as a highway system. Network performance is how fast and smoothly traffic flows. 'Bandwidth' is the number of lanes on the road, while 'latency' is how long it takes a single car to drive from start to finish. Good performance means no traffic jams, no accidents, and your car arriving at its destination quickly and predictably.

22. Describe throughput using a real-life example (BL5)(1)

Answer:
Think of bandwidth as the width of a water pipe, and throughput as the actual amount of water flowing out of the tap per second. Even if you have a large 100 Mbps pipe (bandwidth), if there is a clog or leak (congestion or packet loss), only 20 Mbps of water comes out. That 20 Mbps of delivered water is your throughput.

23. Present how packet loss affects video streaming (BL5)(1)

Answer:
Video streaming is like sending a jigsaw puzzle piece-by-piece over the network. If packet loss happens, some puzzle pieces are lost on the way. A few lost pieces cause the video player to pause or drop quality to re-fetch them (buffering). Heavy packet loss freezes the screen, distorts pixels, or drops the stream because the player cannot reconstruct the picture in real time.

24. Explain delay types clearly in under 1 minute (BL4)(1)

Answer:
Total network delay has four main components: 1. Processing Delay: Time a router takes to inspect a packet header. 2. Queueing Delay: Time the packet waits in line inside the router buffer. 3. Transmission Delay: Time needed to push all packet bits onto the wire. 4. Propagation Delay: Time required for the signal to travel across physical distance at light speed.

25. Describe Cisco Packet Tracer in a professional way (BL5)(1)

Answer:
Cisco Packet Tracer is a visual network simulation tool developed by Cisco. It enables network engineers and students to design, configure, and troubleshoot complex network topologies using virtual routers, switches, firewalls, and end devices in a risk-free environment, allowing real-time and packet-level simulation analysis before physical deployment.

26. Explain QoS in simple and technical terms (BL5)(1)

Answer:
Simple Term: QoS is like an emergency lane on a highway—it lets ambulances (voice and video calls) pass regular traffic so they aren't delayed. Technical Term: QoS is a set of traffic management techniques (DSCP tagging, queuing, policing, shaping) that prioritizes latency-sensitive packets over bulk data to guarantee bounded delay, jitter, and bandwidth.

27. How would you present your simulation results to a client? (BL4)(1)

Answer:
I present simulation results using a top-down approach: First, highlight key outcomes—showing how the network design achieves required speed and availability targets. Second, display clear Packet Tracer topology diagrams. Third, present comparative performance charts showing latency, throughput, and failover recovery times before and after optimization in non-technical terms.

28. Explain jitter using a real-world analogy (BL5)(1)

Answer:
Imagine ordering a package online every day that takes 2 days to arrive. If every package arrives in exactly 2 days, your jitter is zero. But if one package takes 1 day, the next takes 5 days, and another takes 2 days, the arrival timing is unpredictable. That variation in arrival time is jitter. In networking, jitter causes robotic or stuttering audio in voice calls.

29. Describe how you tested network performance in a lab (BL4)(1)

Answer:
In our academic lab, I constructed multi-router topologies in Cisco Packet Tracer. I simulated baseline traffic and generated heavy traffic bursts to induce link congestion. I measured

performance metrics by checking Ping RTT, running `show interfaces` to observe queue depth and dropped packets, and using Simulation Mode to inspect individual PDU processing times across hops.

30. Explain the importance of performance metrics in one structured answer (BL5)(1)

Answer:
Performance metrics are essential because you cannot optimize what you do not measure. Metrics like bandwidth, throughput, delay, jitter, and packet loss provide quantitative visibility into network health. They allow engineers to establish baselines, proactively detect bottlenecks before users are impacted, validate SLA compliance, and justify network upgrades with empirical data.

31. Introduce yourself and explain your knowledge of network performance (BL5)(1)

Answer:
Hello! I am a B.Tech Artificial Intelligence & Data Science student with a strong foundation in Computer Networks. Through my coursework and academic lab work, I have developed an understanding of network protocols, VLAN segmentation, routing, and performance metrics such as throughput, latency, jitter, and packet loss. I have also used Cisco Packet Tracer to design network topologies, configure devices, and observe packet behaviour.

32. Why are you interested in networking and simulation tools? (BL5)(1)

Answer:
I am fascinated by how global communication relies on structured network protocols operating in milliseconds. Networking forms the core infrastructure for digital systems and AI applications. I enjoy using simulation tools like Cisco Packet Tracer because they offer a visual, hands-on environment to design, test, and troubleshoot architectures and observe real-time packet flow without physical risk.

33. What makes you confident in analyzing network performance? (BL5)(1)

Answer:
My confidence comes from a strong grasp of networking fundamentals combined with practical lab experience. I understand how protocol headers, routing lookup, queueing, and physical transmission interact to impact performance. By systematically using diagnostic tools like Packet Tracer simulation mode, ping, and traceroute, I can logically trace bottlenecks to their root cause and recommend fixes.

34. Explain a concept you are very strong in (related to networking) (BL5)(1)

Answer:
I am particularly strong in VLAN Segmentation and Inter-VLAN Routing. VLANs allow us to logically divide a physical switch into isolated broadcast domains, improving network security and reducing broadcast traffic. I understand how to configure 802.1Q trunking links between switches and set up Router-on-a-Stick or Layer 3 switch routing to enable controlled, secure inter-VLAN communication using ACLs.

35. How would you handle a question you don’t know? (BL5)(1)

Answer:
If I encounter a question I don't know, I am honest about not having the immediate answer rather than guessing. I explain what related networking principles I do know and attempt to reason through the problem logically from fundamentals. I then express my eagerness to research the correct solution, learn from the experience, and expand my technical knowledge.

36. Describe a situation where you solved a technical problem (BL4)(1)

Answer:
For example, during a Computer Networks lab exercise in Cisco Packet Tracer, I encountered a situation where devices in one VLAN could not communicate with the default gateway. I approached the problem systematically by checking the IP addresses, subnet masks, interface status, VLAN configuration, and trunk configuration. This type of step-by-step troubleshooting helps identify configuration errors and restore connectivity efficiently.

37. Why should we select you for a networking role? (BL5)(1)

Answer:
You should select me because I bring a solid theoretical and practical foundation in Computer Networks as an AI & Data Science student. I possess strong analytical skills, hands-on experience in network simulation using Cisco Packet Tracer, and a structured approach to troubleshooting. I am a dedicated learner and an effective communicator who can explain technical concepts clearly.

38. What challenges did you face while learning Packet Tracer? (BL5)(1)

Answer:
Initially, my main challenge was understanding how subnets, trunking, and default gateways interacted in multi-VLAN environments, as well as tracking routing convergence. I overcome this by using Packet Tracer's Simulation Mode, which allowed me to step through PDU movements frame-by-frame, observe header changes at each OSI layer, and visually diagnose misconfigurations.

39. How do you improve your technical skills regularly? (BL5)(1)

Answer:
As a student, I regularly improve my technical skills by building and troubleshooting network scenarios in Cisco Packet Tracer, reviewing standard documentation and textbooks, completing lab assignments, and analyzing real-world networking cases. I also practice problem-solving scenarios and reinforce my understanding by explaining technical concepts simply to my peers.

40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

Answer:
I can explain Subnetting simply: Imagine a large apartment building with 256 rooms under one main address. If everyone shouts at once in the main hallway (broadcast traffic), it creates chaos. Subnetting is like building internal floors and doors to separate the building into smaller, quiet sections. It makes communication organized, stops noise from spreading, and ensures only authorized residents enter specific floors, making the network faster and more secure.

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	**30%**	**Demonstrates thorough understanding; answers all questions accurately**	**Shows good understanding with few errors**	**Partial understanding; several incorrect responses**	**Lacks understanding; majority of answers incorrect**
Accuracy of Answers	**25%**	**All answers are correct**	**Most answers are correct with minor mistakes**	**Some correct answers; frequent errors**	**Mostly incorrect answers**
Application of Knowledge	**20%**	**Correctly applies concepts to problem-based questions**	**Applies concepts with minor errors**	**Limited ability to apply concepts**	**Unable to apply concepts**
Time Management	**15%**	**Completes quiz well within time efficiently**	**Completes on time with minor delays**	**Requires extra time or rushes through questions**	**Unable to complete within given time**
Question Interpretation	**10%**	**Interprets all questions correctly and responds appropriately**	**Misinterprets a few questions**	**Often misinterprets questions**	**Unable to understand questions**